YoungJo Choi (Jo Choi)

Jo Choi | Brooklyn, New York, USA | yc7330@nyu.edu | +1-312-918-9643 | **Github**: TechJoe96
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EDUCATION

New York University — Fall 2024-Present
- M.S., Computer Engineering (Expected graduation date: May 2026)

Illinois Institute of Technology (Dual Degree program) — Fall 2022-Spring 2024
- B.S., Computer Engineering, minor in Artificial Intelligence

Inha University (Dual Degree program) — Spring 2016-Spring 2024
- B.S., Electronic Engineering

SKILLS

- **Programming&Scripting:** Python, C++, PostgreSQL, MySQL, C, Java, R, Verilog, SystemVerilog, VHDL, RISC-V Assembly, ARM Assembly, MATLAB
- **AI/ML Frameworks:** PyTorch, TensorFlow, Scikit-learn, Keras, MLflow, Hugging Face
- **Data & Visualization:** Pandas, NumPy, PowerBI, Tableau, Matplotlib, Excel, Selenium
- **Other:** React, HTML/CSS/JavaScript, Node.js, REST API, FastAPI, Flask, SageMaker AWS, Docker, Kubernetes, Linux, Git

WORK EXPERIENCE

Teaching Assistant at New York University — Sep 2025-Dec 2025
- Teaching lab sections for ECE-UY 3114 Fundamentals of Electronics I at New York University
- Ran experiments across amplifiers, diodes and transistor circuits

Data Analyst Intern at **C&B Trading** in Seoul, Korea — Jun 2024-Aug 2024
- Improved sales forecast accuracy to 90% with a predictive model; built Excel reports and PowerBI dashboards that identified three key process bottlenecks for senior management.

Republic of Korea (ROK) Military — Aug 2017-Apr 2019
- Honorably discharged as a Sergeant and a Squad Leader after serving in ROK for 21 months

RESEARCH EXPERIENCE & PROJECTS

Multimodal Gaze Heatmap Prediction from Egocentric Video — Spring 2026

1) Built a compact 3.4M-parameter FiLM-based fusion model in PyTorch (~13 MB, ~2 GFLOPs) combining ResNet-18 vision encoder with audio CNN, trained on 110K+ samples from EGTEA Gaze+
2) Achieved 7.94 px mean gaze prediction error on a 56×56 heatmap grid with ~40ms CPU latency

Efficient Vision-Language Model (VLM) Design with Sparse Techniques — Sep 2025-Present

1) Developing a saliency-based VLM by integrating a Vision Transformer with an LLM through a projection layer, enabling efficient visual token selection and multimodal reasoning under the guidance of Prof. Sai Qian Zhang
2) Reproduced SparseVLM on LLaVA-1.5-7B, reducing visual tokens by 30% with minimal accuracy loss; outperformed three weight-pruning baselines on VQAv2

GenAI-based Chip Design — Fall 2025

1) Developing LLM model architectures for automating chip design across RTL, verification to physical design, also for design optimization under the guidance of Prof. Ramesh Karri
2) Designing a Quantized Neural Net Accelerator with the Microwatt OpenPOWER CPU on SKY130, enabling efficient ML inference via memory-mapped I/O and custom instruction paths

OpenPOWER SoC Design with Microwatt CPU and NN Accelerator — Fall 2025
- FPGA/ASIC-ready neural network accelerator (Verilog RTL) featuring INT8 systolic array processing for OpenPOWER systems

Gender Bias & Rating Prediction in Higher Education Evaluations — Fall 2024
- Analyzed 90K professor records to assess gender bias; built Ridge regression and logistic regression pipelines achieving $R^2 = 0.53$ for rating prediction and ROC AUC = 0.77 for classification

Mouse Motion Tracker using Deep Learning at IIT — Sep 2023-May 2024
- Developed an RNN-based multimodal system using visual and audio data to recognize six mouse behaviors with 90% accuracy under the guidance of Prof. Yan Yan

AWARDS & SCHOLARSHIPS

- Dean's List at IIT — Spring 2023
- Honor Student at Inha University & Merit Scholarship — Fall 2019

LEADERSHIP EXPERIENCE

- Active member at **Triangle fraternity (Armour Chapter)** — Fall 2022-Spring 2024
- **President & Founder** of Korean Student Association at Illinois Institute of Technology — Summer 2023-Spring 2024